CSE4356/CSE5392/EE4328/EE5315 System on Chip
Fall 2025, Lab 1

This lab is due by September 18, 2025. In this lab, you will write Verilog code and configure the Blackboard to implement a 4-nibble hexadecimal counter.

The following steps will guide you through this process:

1. Create a module, hex_to_ss, that converts a 4-bit hexadecimal number to a 8-bit output that is compatible with the format required to driver the cathodes of the seven segment display.
2. Create a module, seven_seg, that accepts a clock input, multidimensional display data (8-bits for each of the 4 seven-segment displays), a 4-bit output to drive the anodes, and an 8-bit output to drive the cathodes. The module should repeatedly cycle through the 4 seven-segment displays driving one anode at a time while time multiplexing the correct display data to the cathodes. The module should cycle through the 4 anodes fast enough to avoid any flicker.
3. Instantiate the module from (2) and instantiate 4 hex_to_ss modules from (1) to drive the display data of the seven_seg module.
4. Add a 16-bit counter that drives the 4 hex_to_ss modules so that the modules output the display data to drive the 4 seven segment displays with the seven_seg module. The counter should initially increment by one every 250ms.
5. Safely condition the PB[0] input and add logic to reset the counter when PB[0] is pressed.
6. Safely condition the SW[11] input and add logic so the counter increments if SW[11] is ‘1’ and decrements if SW[11] is ‘0’ every 250ms.
7. Safely condition SW[10:0] to form a value DELTA. Change the counter to increment or decrement by DELTA every 250 ms.
8. Safely condition the PB[1] and PB[2] inputs. Add logic so that a value MINIMUM is set to the current counter value when PB[1] is pressed. Add logic so that a value MAXIMUM is set to the current counter value when PB[2] is pressed. Add logic so that the default MINIMUM and MAXIMUM are 0x0000 and 0xFFFF on reset. The counter should only count in the range [MINIMUM, MAXIMUM]. Make sure to handle all DELTA values correctly, including a zero value. How to handle the wrap around values when DELTA is not an integer sub-multiple of MAXIMUM-MINIMUM+1 is left for implementer.
9. Safely condition the PB[3] input. Add logic so that a value MATCH is set to the current counter value when PB[3] is pressed.
10. Add logic so that LED[0] toggles when the counter value is equal to MATCH.